Лабораторная №14
Партиции, файловые системы, монтирование

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Цель работы

Получить навыки создания разделов на диске и файловых систем.
Получить навыки монтирования файловых систем.

Рабочая машина

Процессор: Ryzen 7 7435hs 3.10 GHz
Оперативка: DDR5 16x2 4800 MHz
Видеокарта: RTX 4050 6 GB GDDR6

Ход выполнения лабораторной работы

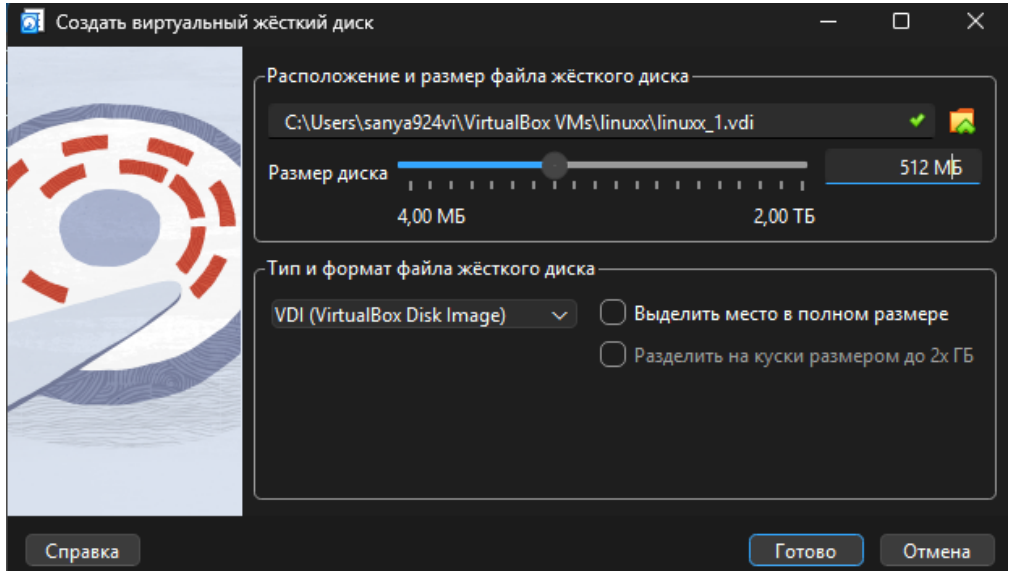

Рис 1. Создаем виртуальный диск

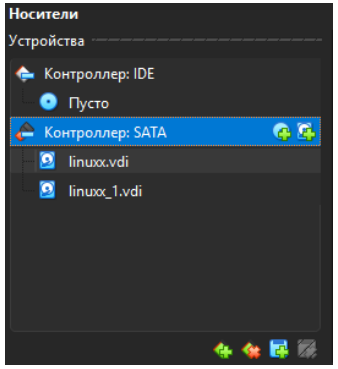

Рис 2. Диск создан

```
adskalozub@localhost:~$ sudo -i
[sudo] пароль для adskalozub:
root@localhost:~# fdisk --list
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes


Disk /dev/sda: 40 GiB, 42949672960 bytes, 83886080 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 1AE3D800-CD7B-437B-B5E6-B804F55A4DB9

Device       Start      End  Sectors  Size Type
/dev/sda1     2048  1230847  1228800  600M EFI System
/dev/sda2  1230848  3327999  2097152    1G Linux extended boot
/dev/sda3  3328000 83884031 80556032 38,4G Linux LVM
```

Рис 3. Проверка дисков

```
root@localhost:~# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xe21aa5d4.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

root@localhost:~# m
bash: m: команда не найдена...
root@localhost:~# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): m

Help:

  DOS (MBR)
   a   toggle a bootable flag
   b   edit nested BSD disklabel
   c   toggle the dos compatibility flag
```

Рис 4. Проверка инструментов

```
Command (m for help): p
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xe21aa5d4
```

Рис 5. Вывод информации

```
Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-1048575, default 2048): 2048
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-1048575, default 1048575): +100
M

Created a new partition 1 of type 'Linux' and of size 100 MiB.
```

Рис 6. Создаем раздел размером в 100 Мбайт

```
Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all):
Hex code or alias (type L to list all): 83
Changed type of partition 'Linux' to 'Linux'.
```

Рис 7. изменяем раздел

```
root@localhost:~# fdisk -l /dev/sdb
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xe21aa5d4
root@localhost:~# cat /proc/partitions
major minor  #blocks  name

   8       16     524288 sdb
   8        0   41943040 sda
   8        1     614400 sda1
   8        2    1048576 sda2
   8        3   40278016 sda3
  11        0    1048575 sr0
 253        0   36081664 dm-0
 253        1    4194304 dm-1
root@localhost:~# partprobe /dev/sdb
```

Рис 8. Проверка

```
Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): e
Partition number (1-4, default 1): 4
First sector (2048-1048575, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-1048575, default 1048575):

Created a new partition 4 of type 'Extended' and of size 511 MiB.
```

Рис 9. Создаем новый раздел

```
Command (m for help): n
All space for primary partitions is in use.
Adding logical partition 5
First sector (4096-1048575, default 4096):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (4096-1048575, default 1048575): +101
M

Created a new partition 5 of type 'Linux' and of size 101 MiB.
```

Рис 10. Создаем новый раздел

```
root@localhost:~# partprobe /dev/sdb
root@localhost:~# cat /proc/partitions
major minor  #blocks  name

   8       16     524288 sdb
   8       20          0 sdb4
   8       21     103424 sdb5
   8        0   41943040 sda
   8        1     614400 sda1
   8        2    1048576 sda2
   8        3   40278016 sda3
  11        0    1048575 sr0
 253        0   36081664 dm-0
 253        1    4194304 dm-1
root@localhost:~# fdisk --list /dev/sdb
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xe21aa5d4

Device     Boot Start     End Sectors  Size Id Type
/dev/sdb4        2048 1048575 1046528  511M  5 Extended
/dev/sdb5        4096  210943  206848  101M 83 Linux
```

Рис 11. проверка

```
root@localhost:~# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): n
All space for primary partitions is in use.
Adding logical partition 6
First sector (212992-1048575, default 212992):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (212992-1048575, default 1048575): +1
00M

Created a new partition 6 of type 'Linux' and of size 100 MiB.
```

Рис 12. создание раздела

```
Command (m for help): t
Partition number (4-6, default 6):
Hex code or alias (type L to list all): 82

Changed type of partition 'Linux' to 'Linux swap / Solaris'.
```

Рис 13. Выбираем раздел

```
root@localhost:~# partprobe /dev/sdb
root@localhost:~# cat /proc/partitions
major minor  #blocks  name

   8       16     524288 sdb
   8       20          0 sdb4
   8       21     103424 sdb5
   8       22     102400 sdb6
   8        0   41943040 sda
   8        1     614400 sda1
   8        2    1048576 sda2
   8        3   40278016 sda3
  11        0    1048575 sr0
 253        0   36081664 dm-0
 253        1    4194304 dm-1
root@localhost:~# fdisk --list /dev/sdb
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xe21aa5d4

Device     Boot  Start     End Sectors  Size Id Type
/dev/sdb4         2048 1048575 1046528  511M  5 Extended
/dev/sdb5         4096  210943  206848  101M 83 Linux
/dev/sdb6       212992  417791  204800  100M 82 Linux swap / Solaris
root@localhost:~# mkswap /dev/sdb6
Setting up swapspace version 1, size = 100 MiB (104853504 bytes)
no label, UUID=8e30dac3-1d4a-4b41-b217-7dae55fd6b29
root@localhost:~# swapon /dev/sdb6
root@localhost:~# free -m
               total        used        free      shared  buff/cache   available
Mem:            7660        2182        4336          99        1477        5477
Swap:           4195           0        4195
```

Рис 14. Проверка

```
root@localhost:~# gdisk -l /dev/sdc
bash: gdisk: команда не найдена...
```

Рис 15. Не удалось выполнить этап из-за Технической несостыковки

```
root@localhost:~# mkfs.xfs /dev/sdb1
Error accessing specified device /dev/sdb1: Нет такого файла или каталога
Usage: mkfs.xfs
/* blocksize */          [-b size=num]
/* config file */        [-c options=xxx]
/* metadata */           [-m crc=0|1,finobt=0|1,uuid=xxx,rmapbt=0|1,reflink=0|1,
                            inobtcount=0|1,bigtime=0|1,autofsck=xxx]
/* data subvol */        [-d agcount=n,agsize=n,file,name=xxx,size=num,
                            (sunit=value,swidth=value|su=num,sw=num|noalign),
                            sectsize=num,concurrency=num]
/* force overwrite */    [-f]
/* inode size */         [-i perblock=n|size=num,maxpct=n,attr=0|1|2,
                            projid32bit=0|1,sparse=0|1,nrext64=0|1,
                            exchange=0|1]
/* no discard */         [-K]
/* log subvol */         [-l agnum=n,internal,size=num,logdev=xxx,version=n
                            sunit=value|su=num,sectsize=num,lazy-count=0|1,
                            concurrency=num]
/* label */              [-L label (maximum 12 characters)]
/* naming */             [-n size=num,version=2|ci,ftype=0|1,parent=0|1]]
/* no-op info only */    [-N]
/* prototype file */     [-p fname]
/* quiet */              [-q]
/* realtime subvol */    [-r extsize=num,size=num,rtdev=xxx]
/* sectorsize */         [-s size=num]
/* version */            [-V]
                         devicename
<devicename> is required unless -d name=xxx is given.
<num> is xxx (bytes), xxxs (sectors), xxxb (fs blocks), xxxk (xxx KiB),
      xxxm (xxx MiB), xxxg (xxx GiB), xxxt (xxx TiB) or xxxp (xxx PiB).
<value> is xxx (512 byte blocks).
root@localhost:~# xfs_admin -L xfsdisk /dev/sdb1
/dev/sdb1: Нет такого файла или каталога

fatal error -- couldn't initialize XFS library
```

Рис 16. вывод информации

```
root@localhost:~# mkfs.ext4 /dev/sdb5
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 103424 1k blocks and 25896 inodes
Filesystem UUID: 0e694203-b227-4f5f-a1a0-2a1595d727e4
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

root@localhost:~# tune2fs -L ext4disk /dev/sdb5
tune2fs 1.47.1 (20-May-2024)
root@localhost:~# tune2fs -o acl,user_xattr /dev/sdb5
tune2fs 1.47.1 (20-May-2024)
```

Рис 16. вывод информации

```
root@localhost:~# mkdir -p /mnt/tmp
root@localhost:~# mount /dev/sdb5 /mnt/tmp
root@localhost:~# mount
/dev/mapper/rl-root on / type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,log
bsize=32k,noquota)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=3894340k,nr_inodes=973585,m
ode=755,inode64)
```

Рис 17. Настройка раздела и проверка

```
root@localhost:~# umount /dev/sdb5
root@localhost:~# mount
/dev/mapper/rl-root on / type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,log
bsize=32k,noquota)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=3894340k,nr_inodes=973585,m
ode=755,inode64)
```

Рис 18. Настройка и проверка

```
root@localhost:~# mkdir -p /mnt/data
root@localhost:~# blkid
/dev/mapper/rl-root: UUID="817554fb-cd64-49be-a809-40bc3d466a69" BLOCK_SIZE="512" T
YPE="xfs"
/dev/sda3: UUID="c1n6CR-mvxb-3xtY-48dj-fyfj-v641-uiiBrY" TYPE="LVM2_member" PARTUUI
D="9ecc4106-f8ed-425f-87bb-897416b39054"
/dev/mapper/rl-swap: UUID="b4d4013d-cadd-4b6e-931a-b95022f1bf32" TYPE="swap"
/dev/sdb4: PTTYPE="dos" PARTUUID="e21aa5d4-04"
/dev/sdb5: LABEL="ext4disk" UUID="0e694203-b227-4f5f-a1a0-2a1595d727e4" BLOCK_SIZE=
"1024" TYPE="ext4" PARTUUID="e21aa5d4-05"
/dev/sdb6: UUID="8e30dac3-1d4a-4b41-b217-7dae55fd6b29" TYPE="swap" PARTUUID="e21aa5
d4-06"
/dev/sda2: UUID="830074e3-7303-41c9-9e1a-966908e9dfb7" BLOCK_SIZE="512" TYPE="xfs"
PARTUUID="04832f16-d603-4253-a34b-0a4a7e2dd9be"
/dev/sda1: UUID="EF67-AE5A" BLOCK_SIZE="512" TYPE="vfat" PARTLABEL="EFI System Part
ition" PARTUUID="ee6403b4-3a50-4a4d-841e-ed71f60f8435"
root@localhost:~# blkid /dev/sdb1
```

Рис 19. Создаем директорию и выводим информацию

```
root@localhost:~# mount -a
root@localhost:~# df -h
Файловая система     Размер Использовано  Дост Использовано% Смонтировано в
/dev/mapper/rl-root    35G         5,1G   30G           15% /
devtmpfs              3,8G            0  3,8G            0% /dev
tmpfs                 3,8G          84K  3,8G            1% /dev/shm
efivarfs              256K          51K  201K           20% /sys/firmware/efi/efiva
rs
tmpfs                 1,5G         1,3M  1,5G            1% /run
tmpfs                 1,0M            0  1,0M            0% /run/credentials/system
d-journald.service
/dev/sda2             960M         459M  502M           48% /boot
/dev/sda1             599M         8,4M  591M            2% /boot/efi
tmpfs                 767M         148K  766M            1% /run/user/1000
tmpfs                 767M          56K  766M            1% /run/user/0
```

Рис 20. Выводим информацию по разделам

Вывод

Научились создавать и редактировать разделы дисков